

Little Red Dots: Super-Eddington Winds, Not Overmassive Black Holes

Type IIn supernovae and η Carinae's Great Eruption reveal a hidden engine scenario for JWST's enigmatic Little Red Dots.

arXiv:2606.30711 — Little Red Dots as intermediate-mass, super-Eddington engines.

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Background

JWST's Little Red Dots show blackbody-like continua, broad lines, and Balmer breaks unlike any known galaxy or AGN. Similar features appear in Type IIn supernovae and η Carinae's 19th-century Great Eruption.

Are Little Red Dots overmassive black holes or something else entirely?

What this paper is selling

The authors propose that dense, slow winds from an intermediate-mass engine create a pseudo-photosphere, mimicking a virial broad-line region. This wind physics may explain the entire LRD life cycle, from enshrouded youth to massive black hole seeds.

Key results to remember

- Engine mass $< 10^5$ solar masses, likely 10^3 – 6 , super-Eddington by factor ≥ 5 .
- Dense wind pseudo-photosphere yields blackbody continuum and broad lines.
- Dust production in envelope may mark the birth of classical AGN.

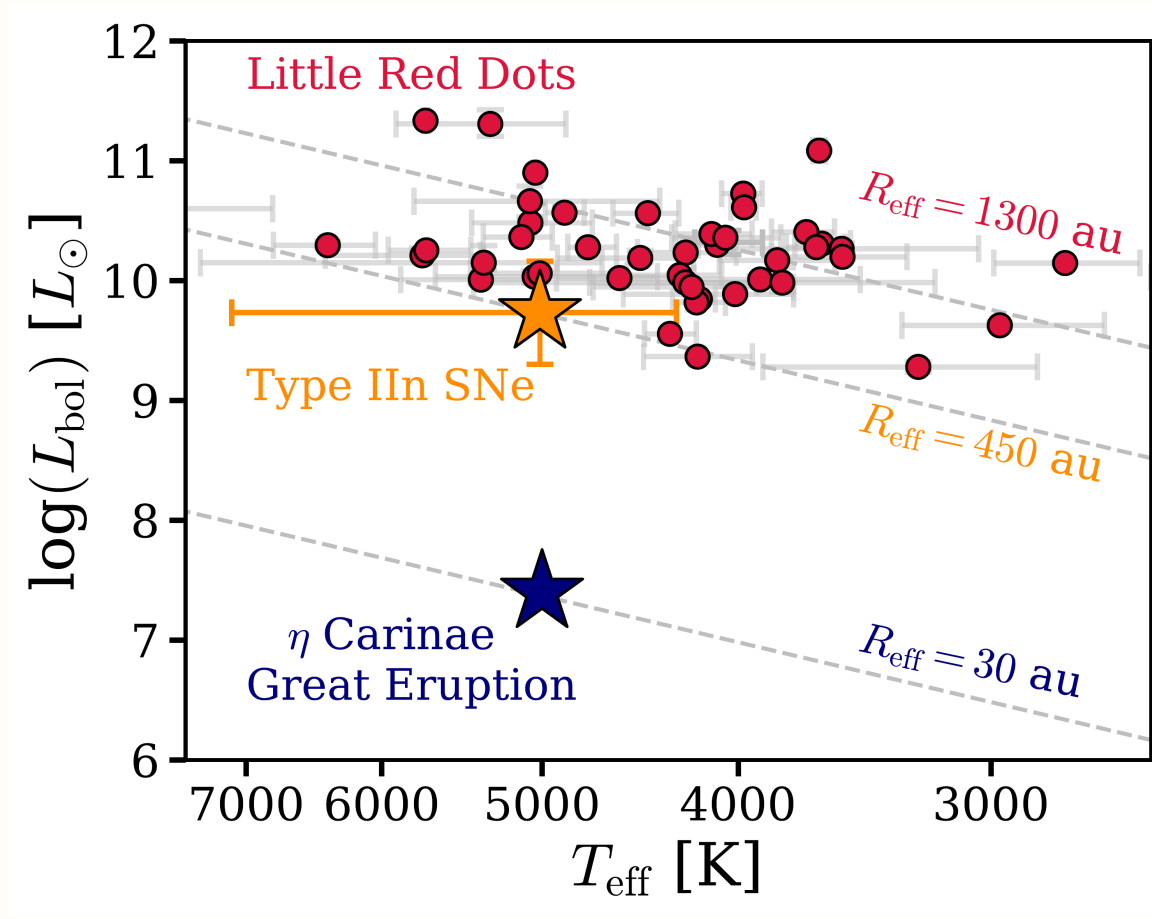


Fig. 4. LRDs, Type IIn SNe, and η Carinae cluster at similar effective temperatures despite spanning four decades in luminosity, revealing a common pseudo-photosphere.

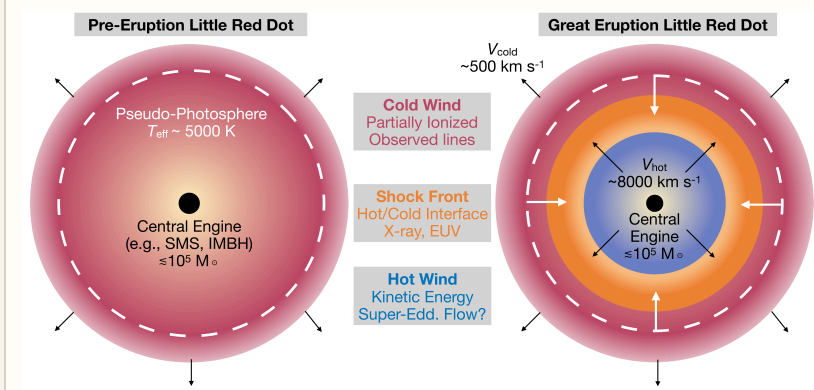


Fig. 8. Schematic shows how a slow cold wind enshrouds the engine, then fast winds crash into it, producing shocks that power the LRD.